

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q19-1325

Fountain City 138 kV
July 15, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1325. The Interconnection Customer proposes a 131.9 MW solar photovoltaic facility interconnecting at the Fountain City 138 kV substation in the Maryland area. The request was received on October 29, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 383 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,412,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Bruce Crossing - Holopaw Substation 138.0 kV line	Sierra Pacific Lumber 138-1 kV Transformer Bank	111.2	290	8.1
Rio Blanco 2702 - Alliance 138.0 kV L	Sierrapacific Lumber 138/1 kV Transformer	111.2	247	8.1
Rio Blanco 2702 Substation - Bruce Crossng 138.0 kV	Sierra Pacific Lumber 138/1 kV TX	101.8	294	8.1

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Bruce Crossing - Holopaw Substation 138.0 kV line	Rebuild line with higher-capacity conductor	\$7,830,000	15% (\$1,148,000)
Rio Blanco 2702 - Alliance 138.0 kV L	Rebuild line with higher-capacity conductor	\$1,024,000	15% (\$150,000)
Bruce Crossing - Rio Blanco 2702 138.0 kV	Reconductor line	\$775,000	15% (\$114,000)
	Total allocated to project		\$1,412,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.